One-dimensional arrays

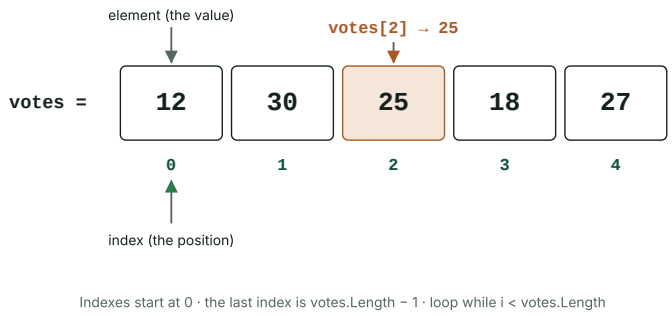
D17 FACT FILE

Specification: students should be able to

- One-dimensional array structures

CCEA fact file: Programming Constructs 1

Key words: array, list, element, index, zero-indexed, initialise, traverse, length



An array of five votes showing elements and zero-based indexes

Arrays

A one-dimensional array stores a collection of values of the same data type under a single identifier. Each element is accessed by its index, which starts at 0. Arrays allow loops to process many values with the same few lines of code. Elements can be read, changed and processed; the index identifies the position, not the value.

Processing arrays

Use a FOR loop over the indexes (for (int i = 0; i < arr.Length; i++)) or a WHILE loop with a counter that increases each time. Typical tasks: set all elements to zero, total all elements, count elements that meet a condition, and find the minimum or maximum by keeping a 'best so far' value.

T Key terms from the mark schemes

Array
A data structure holding values of one data type under one name, accessed by index.
Credited: U4 2024 Q7, 2025 Q8

Index
The position number of an element in an array (starting at 0).
Credited: U4 2025 Q8

C C# example

Creating and accessing

```
int[] votes = { 12, 30, 25, 18, 27 };   // 5 elements, indexes 0-4
Console.WriteLine(votes[0]);      // 12 — first element
Console.WriteLine(votes[4]);      // 27 — last element
votes[2] = 26;                    // update the third element
Console.WriteLine(votes.Length);  // 5

string[] names = new string[3];   // empty array of 3 strings
names[0] = "Ali";
```

Exam focus

Array questions are worth 15–25 marks: 2022 Q7, 2023 Q9, 2024 Q7, 2025 Q8. Common errors: not using a loop when asked, not using the array name given, using the wrong index, placing outputs inside the loop. Know that arrays are zero-indexed and that their contents can be changed (2025 Q8(a)).

Unit 4 2025 Q8	[20]	Array properties, initialise, count, WHILE totals
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q7	[18]	Array indexing; weekly totals and average algorithm; functions and their benefits

? Check yourself

1. Describe what is meant by an array. [2]
2. An array votes holds [14, 22, 9, 31]. What is the value of votes[2]? [1]
3. Write a statement that changes the fifth element of the array heights to 162. [2]

Answers: 1. A data structure that stores many values of the same data type under one name/identifier (1) / each value (element) is accessed by its index, starting at 0 (1) 2. 9 (1) 3. heights[4] ← 162 (C#: heights[4] = 162;). Award: index 4 (zero-indexed fifth element) (1) / correct assignment of 162 to heights (1)